

Question ID: 8637294f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

If $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$, what is the value of n ?

Correct Answer: .0625, 1/16

Rationale

The correct answer is **.0625**. It's given that $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$. The equation $\frac{4a}{b} = 6.7$ can be rewritten as $(4)(\frac{a}{b}) = 6.7$. Dividing both sides of this equation by 4 yields $\frac{a}{b} = 1.675$. The equation $\frac{a}{bn} = 26.8$ can be rewritten as $(\frac{a}{b})(\frac{1}{n}) = 26.8$. Substituting **1.675** for $\frac{a}{b}$ in this equation yields $(1.675)(\frac{1}{n}) = 26.8$, or $\frac{1.675}{n} = 26.8$. Multiplying both sides of this equation by n yields **1.675 = 26.8n**. Dividing both sides of this equation by **26.8** yields $n = 0.0625$. Therefore, the value of n is **0.0625**. Note that .0625, 0.062, 0.063, and 1/16 are examples of ways to enter a correct answer.